Hikvision Real-Time Alerts Integration (Backend + Frontend)

Architecture Overview

Hikvision Camera/NVR → Backend Server (Node.js or FastAPI) → WebSocket → React Frontend Dashboard

1. Node.js Backend (Real-Time Alert Listener)

```
const http = require("http");
const express = require("express");
const { Server } = require("socket.io");

const app = express();
const server = http.createServer(app);
const io = new Server(server, { cors: { origin: "*" } });

const options = {
  hostname: "192.168.1.64",
  port: 80,
  path: "/ISAPI/Event/notification/alertStream",
  auth: "admin:password",
  headers: { Connection: "keep-alive" }
};

function connectStream() {
  http.get(options, res => {
    res.on("data", chunk => {
      const data = chunk.toString();
      if (data.includes("<EventNotificationAlert")) {
        io.emit("hikvision-alert", {
          message: "Motion Detected",
          raw: data
        });
      }
    });
    res.on("end", () => setTimeout(connectStream, 5000));
  }).on("error", () => setTimeout(connectStream, 5000));
}

connectStream();
server.listen(5000, () => console.log("Server running on port 5000"));
```

2. Python FastAPI Backend (Alternative)

```
import requests
from fastapi import FastAPI, WebSocket
import asyncio
import threading

app = FastAPI()
clients = []

@app.websocket("/ws/alerts")
async def websocket_endpoint(websocket: WebSocket):
    await websocket.accept()
    clients.append(websocket)
    try:
```

```
        while True:
            await asyncio.sleep(1)
    except:
        clients.remove(websocket)

async def broadcast_alert(message):
    for ws in clients:
        await ws.send_text(message)

def listen_to_hikvision():
    url = "http://192.168.1.64/ISAPI/Event/notification/alertStream"
    auth = ("admin", "password")
    with requests.get(url, auth=auth, stream=True) as r:
        for chunk in r.iter_content(chunk_size=1024):
            if b"EventNotificationAlert" in chunk:
                asyncio.run(broadcast_alert("Motion Detected"))

threading.Thread(target=listen_to_hikvision, daemon=True).start()
```

3. React Frontend (Live Alerts UI)

```
import { useEffect, useState } from "react";
import { io } from "socket.io-client";

const socket = io("http://localhost:5000");

export default function HikvisionAlerts() {
  const [alerts, setAlerts] = useState([]);

  useEffect(() => {
    socket.on("hikvision-alert", (alert) => {
      setAlerts(prev => [alert, ...prev]);
    });
    return () => socket.off("hikvision-alert");
  }, []);

  return (
    <div>
      <h2>Live Camera Alerts</h2>
      {alerts.map((a, i) => (
        <div key={i} style={{background:"#ffe6e6", margin:"10px", padding:"10px"}}>
          {a.message}
        </div>
      ))}
    </div>
  );
}
```

This completes the full stack flow from Hikvision device alerts to a live React dashboard.